

Limits of other indeterminate forms — 0.4.8

$\infty - \infty$: often can combine into one function

Example: $\lim_{x \rightarrow 0} \frac{1}{x^2} - \frac{1}{x^4} = \lim_{x \rightarrow 0} \frac{x^2 - 1}{x^4} \begin{matrix} \nearrow -1 \\ \searrow 0^+ \end{matrix} = -\infty$

$0 \cdot \infty$: can rewrite as a quotient

Example: $\lim_{x \rightarrow 0^+} \overset{0}{x} \overset{\infty}{e^{1/x}} = \lim_{x \rightarrow 0^+} \frac{e^{1/x}}{1/x} \begin{matrix} \nearrow \infty \\ \searrow \infty \end{matrix}$

L'Hôpital $\lim_{x \rightarrow 0^+} \frac{e^{1/x} \cdot \cancel{\frac{-1}{x^2}}}{\cancel{\frac{-1}{x^2}}} = \lim_{x \rightarrow 0^+} e^{1/x} = \infty$

Example : $\lim_{x \rightarrow \frac{\pi}{2}} (x - \frac{\pi}{2}) \tan(x)$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{x - \frac{\pi}{2}}{\cot(x)} \quad \text{L'H} \quad = \lim_{x \rightarrow \frac{\pi}{2}} \frac{1}{-\csc^2(x)} = \lim_{x \rightarrow \frac{\pi}{2}} -\sin^2(x) = -1$$

Generally $0 \cdot \infty = \frac{0}{\frac{1}{\infty}} = \frac{\infty}{\frac{1}{0}}$

Other indeterminate forms: 0^0 , 1^∞ , ∞^0

Idea: use logarithms to turn exponentiation into multiplication!

$$\lim_{x \rightarrow 0} 0^x = 0, \quad \lim_{x \rightarrow 0} x^0 = 1$$

Example: what is $\lim_{x \rightarrow 0^+} x^x$? $\lim_{x \rightarrow 0^+} x^x = L$

$$\ln(L) = \ln\left(\lim_{x \rightarrow 0^+} x^x\right) = \lim_{x \rightarrow 0^+} \ln(x^x) = \lim_{x \rightarrow 0^+} x \overset{\nearrow 0}{\ln(x)} \overset{\nearrow -\infty}{} =$$

$$= \lim_{x \rightarrow 0^+} \frac{\overset{\nearrow \infty}{\ln(x)}}{\overset{\nearrow \infty}{1/x}} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2}$$

$$= \lim_{x \rightarrow 0^+} -\frac{x^2}{x} = \lim_{x \rightarrow 0^+} -x = 0$$

Not the final answer! We've computed that $\ln(L) = 0$, so we want to apply the inverse function e^x to both sides.

$$L = e^{\ln(L)} = e^0 = 1.$$

$$\boxed{\lim_{x \rightarrow 0^+} x^x = 1}$$

Example: $\lim_{x \rightarrow \infty} x^{1/x}$ is indeterminate of type ∞^0

Suppose $L = \lim_{x \rightarrow \infty} x^{1/x} \Rightarrow \ln(L) = \lim_{x \rightarrow \infty} \ln(x^{1/x})$

$$= \lim_{x \rightarrow \infty} \frac{1}{x} \cdot \ln(x) = \lim_{x \rightarrow \infty} \frac{\ln(x) \xrightarrow{\infty}}{x \xrightarrow{\infty}} \stackrel{L'H}{=} \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$$

$$\ln(L) = 0 \Rightarrow L = e^0 = 1.$$

$$\lim_{x \rightarrow 0^+} \frac{\cos(x) \xrightarrow{1}}{\sin(x) \xrightarrow{0^+}} = \infty$$

Example: $\lim_{x \rightarrow 0^+} (1+\sqrt{x})^{\cot(x)}$ indeterminate of type 1^∞

$$L = \lim_{x \rightarrow 0^+} (1+\sqrt{x})^{\cot(x)} \Rightarrow \ln(L) = \lim_{x \rightarrow 0^+} \cot(x) \ln(1+\sqrt{x})$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln(1+\sqrt{x}) \xrightarrow{0}}{\tan(x) \xrightarrow{0}} \stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \frac{\frac{1}{1+\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}}{\sec^2(x)}$$

$$= \lim_{x \rightarrow 0^+} \frac{\cos^2(x) \xrightarrow{1}}{(1+\sqrt{x}) \cdot 2\sqrt{x} \xrightarrow{0^+}} = \infty$$

$$\ln(L) = \infty$$

$$L = \lim_{x \rightarrow \infty} e^x = \infty$$

$$"L = e^\infty = \infty"$$